

Standard quantum mechanics

Cauchy (initial-value) problem

$\psi(x, t_0)$ given $\rightarrow$ evolve forward

$$i\hbar \partial_t \psi = H\psi$$

Schrodinger / Dirac equation

$$|\psi|^2 = \text{Born rule}$$

Fundamental axiom

Time: fundamental coordinate

Lorentzian spacetime a priori

Measurement: extra postulate

Collapse / decoherence added

Euclidean Condensate Theory

Boundary-value (elliptic) problem

$\Phi(X)$ on entire M^4

$$\delta A B \partial A \partial B \Phi - V'(\Phi) = 0$$

Euclidean condensate equation

$$|\psi|^2 \text{ from coarse-graining}$$

Emergent, not axiom

Time: emergent from SSB

$O(4) \rightarrow O(3)$ selects w-direction

Measurement: $E \rightarrow L$ transition

Under environmental locking